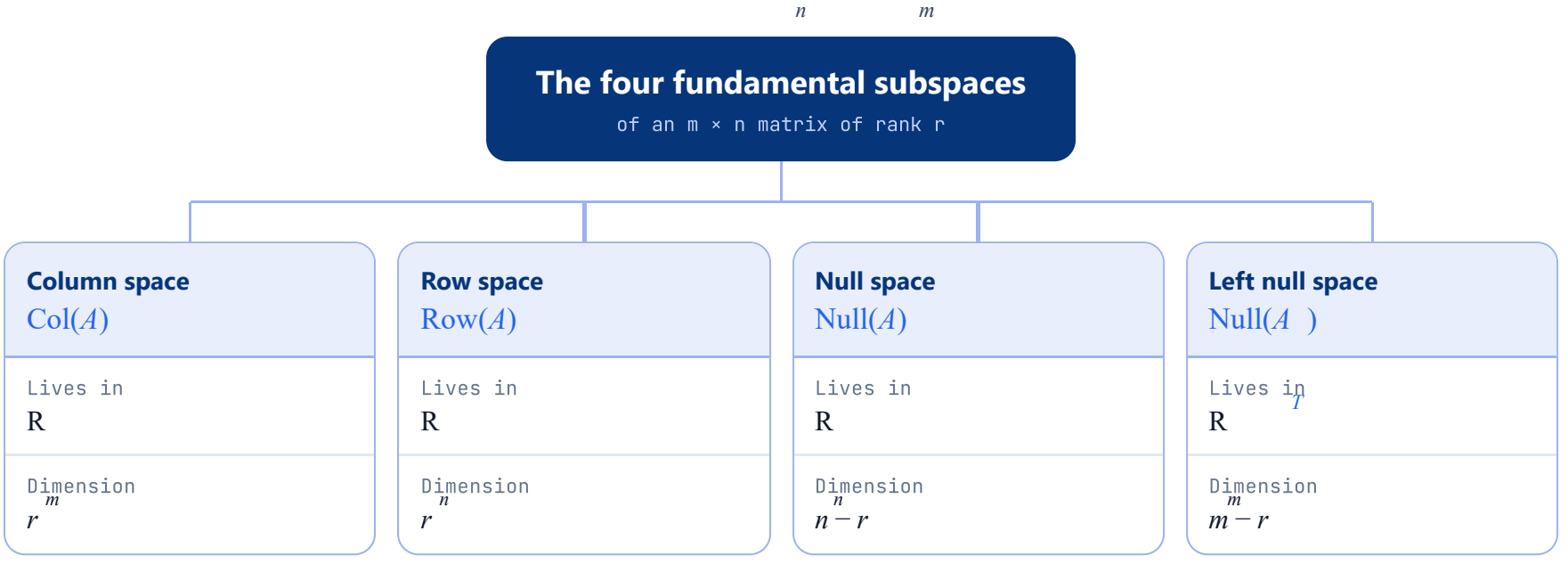
BEFORE YOU READ

Four subspaces from one matrix

Every $m \times n$ matrix A of rank r carries four subspaces: two live in $\mathbb{R}$, two in $\mathbb{R}$.



TAKEAWAY

They come in complementary pairs

The row space and the null space of A complete each other in $\mathbb{R}$; the column space and the null space of A complete each other in $\mathbb{R}$.

n T

m